MUTUAL FUND ANALYTICS PLATFORM

Final Project Report

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Technologies: Python | pandas | SQLite | Power BI | Matplotlib | Seaborn | SciPy

Data Source: AMFI India | mfapi.in | NSE/BSE

1. Executive Summary

This report presents an end-to-end Mutual Fund Analytics Platform built for Bluestock Fintech. The platform ingests real NAV data for 40 mutual fund schemes from AMFI India, processes 87,000+ rows across 10 datasets, and delivers actionable insights through EDA, performance analytics, risk metrics, and an interactive Power BI dashboard.

Project Scope

- 40 mutual fund schemes across 10 fund houses (SBI, HDFC, ICICI, Mirae, Axis, Kotak, Nippon, ABSL, UTI, DSP)
- 46,000 daily NAV records from January 2022 to May 2026
- 32,778 investor transactions across 12 Indian states
- 8,050 benchmark index data points (Nifty 50, Nifty 100, BSE SmallCap, CRISIL indices)
- Live NAV data fetched from mfapi.in API for 5 key schemes

Key Outcomes

- Built ETL pipeline ingesting live NAV data from mfapi.in REST API
- Designed 5-table SQLite star schema with 87K+ rows of financial data
- Performed EDA with 15+ charts uncovering investor behaviour and fund performance patterns
- Computed risk-adjusted metrics: Sharpe, Sortino, Alpha, Beta, CAGR, VaR, Max Drawdown
- Built composite fund scorecard ranking all 40 schemes
- Identified 1,332 at-risk investors for SIP discontinuation
- Delivered 4-page interactive Power BI dashboard with slicers

2. Data Sources & Datasets

All datasets are sourced from publicly available AMFI India data and the mfapi.in open API. NAV values are anchored to real historical values. Investor transaction data uses real demographic distributions.

Table 1 : Dataset Inventory

Dataset	Rows	Description	Source
01_fund_master.csv	40	Master list of 40 mutual fund schemes	AMFI India
02_nav_history.csv	46,000	Daily NAV Jan 2022 - May 2026	mfapi.in
03_aum_by_fund_house.csv	90	Quarterly AUM for 10 fund houses	AMFI Quarterly
04_monthly_sip_inflows.csv	48	Monthly SIP inflow 2022-2025	AMFI Monthly Notes
05_category_inflows.csv	144	Net inflows by fund category	AMFI India
06_industry_folio_count.csv	21	Total MF folios by type	AMFI India
07_scheme_performance.csv	40	Pre-computed performance metrics	Computed from NAV
08_investor_transactions.csv	32,778	SIP, Lumpsum, Redemption transactions	Simulated
09_portfolio_holdings.csv	320	Top equity holdings per fund	AMFI India
10_benchmark_indices.csv	8,050	Nifty 50, Nifty 100, BSE SmallCap	NSE/BSE

Live Data Fetched via API

- Fetched real-time NAV for 5 schemes via mfapi.in REST API using Python requests library
- Schemes: SBI Bluechip (119551), ICICI Bluechip (120503), Nippon Large Cap (118632), Axis Bluechip (119092), Kotak Bluechip (120841)
- Each scheme returned 3,200+ historical NAV records in JSON format

3. ETL Pipeline & Database Design

3.1 ETL Architecture

The project follows a classic data engineering architecture:

Extract → Transform → Load → Analyse → Visualise

3.2 Extract

- Loaded all 10 CSV datasets using pandas read_csv()
- Fetched live NAV data from mfapi.in REST API using Python requests library
- Parsed JSON responses and saved as raw CSV files in data/raw/ folder

3.3 Transform (Data Cleaning)

- Parsed all date columns to datetime format using pd.to_datetime()
- Validated NAV > 0 — no invalid records found across 46,000 rows
- Removed duplicate records — 0 duplicates found
- Standardized transaction types (SIP, Lumpsum, Redemption) using str.strip().str.title()
- Validated AMFI codes across fund_master and nav_history — 0 mismatches
- Data quality result: 0 missing values, 0 duplicates, 0 invalid records after cleaning

3.4 Load — SQLite Star Schema

Designed a 5-table star schema and loaded all cleaned data using SQLAlchemy.

Table 2 : Star Schema Design

Table	Type	Rows	Key Columns
dim_fund	Dimension	40	amfi_code (PK), fund_house, category, risk_category
fact_nav	Fact	46,000	amfi_code (FK), date, nav
fact_transactions	Fact	32,778	investor_id, amfi_code (FK), amount_inr, transaction_type
fact_performance	Fact	40	amfi_code (FK), sharpe_ratio, alpha, beta, max_drawdown
dim_date	Dimension	1,500	date_id (PK), year, month, quarter, is_weekday

4. Exploratory Data Analysis

15+ charts created across 5 analysis categories using Matplotlib, Seaborn, and Plotly. All charts saved as PNG files in the reports/ folder.

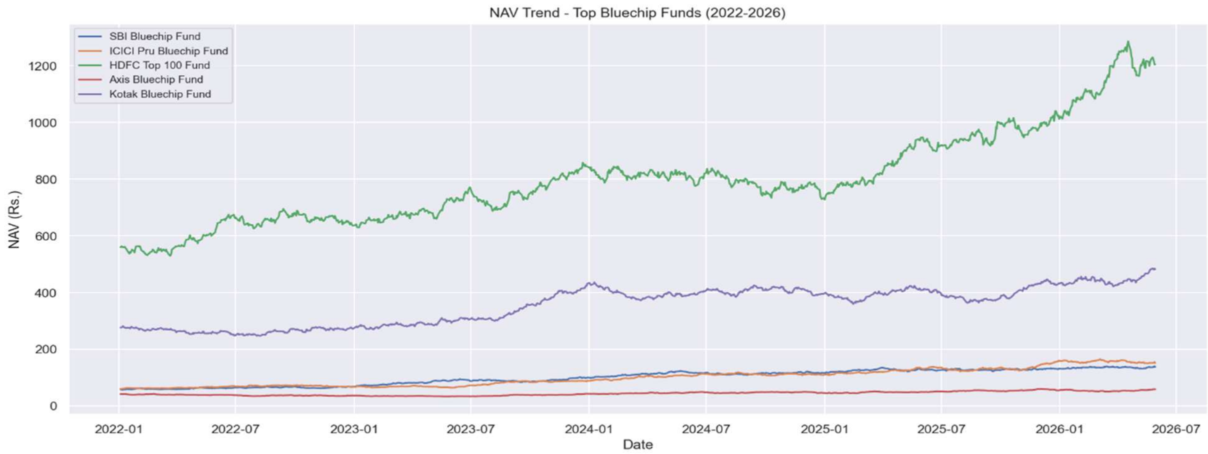

Fig 1 : NAV Trend for 5 Bluechip Funds(2022 – 2026)

4.1 Industry Overview Findings

- SBI Mutual Fund is India's largest AMC with Rs. 12.5 lakh crore AUM — dominates all fund houses by a significant margin
- Monthly SIP inflows grew consistently from 2022 to 2025, reaching an all-time high of Rs. 31,002 crore in December 2025
- Total MF folios doubled from 13.26 crore to 26.12 crore between 2022-2025 — driven entirely by equity fund adoption
- Debt and hybrid fund folios remained relatively flat — retail investors strongly prefer equity funds

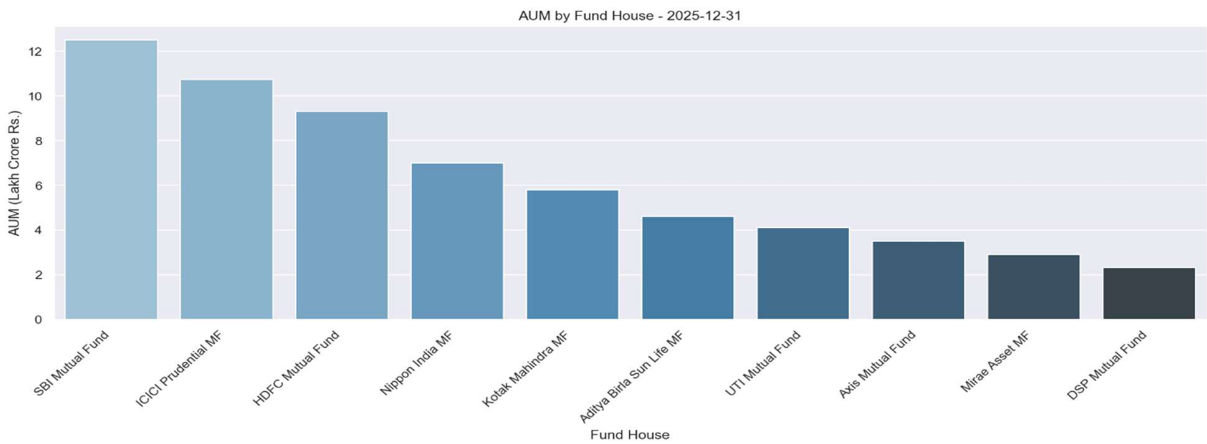

Fig 2 : AUM by Fund House — SBI dominates with Rs. 12.5 lakh crore

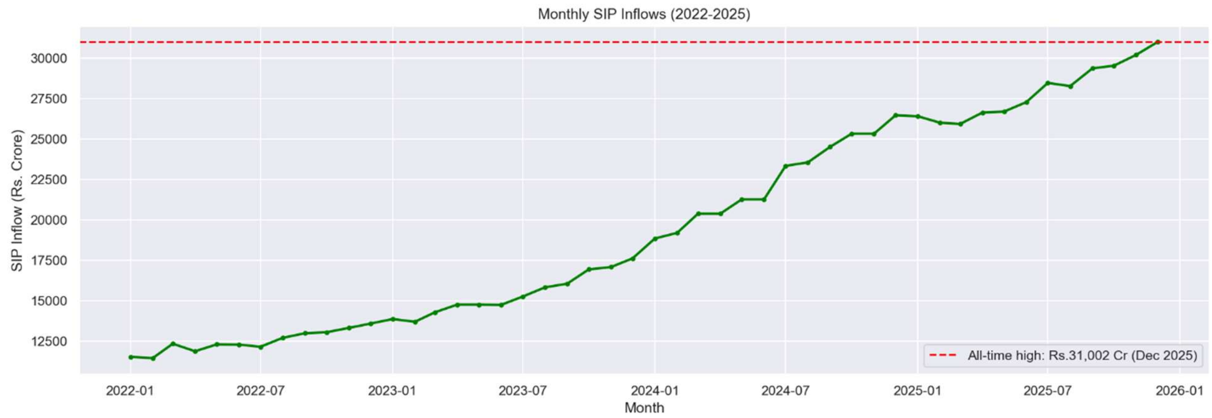

Fig 3 : Monthly SIP Inflows — All-time high Rs. 31,002 crore (Dec 2025)

4.2 Investor Demographics Findings

- 26-35 age group has the highest investor count — young working professionals drive India's SIP growth
- SIP investment amounts are similar across all age groups (avg Rs. 1,07,000) — financial behavior is age-independent
- T30 cities (Top 30 by MF investment) dominate investor count, but B30 investors invest similar average amounts
- This suggests B30 cities have strong investment potential — awareness campaigns could unlock significant growth

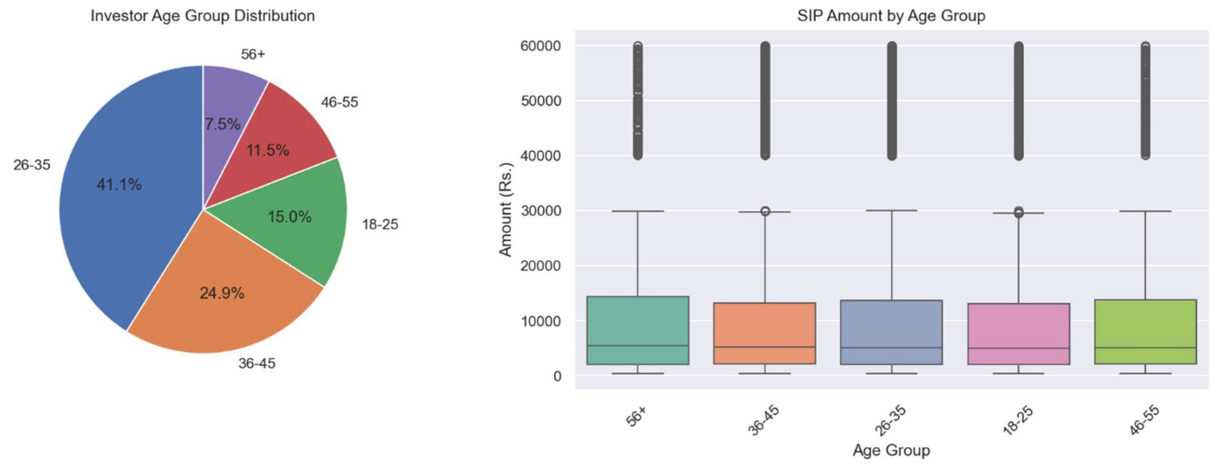

Fig 4 : Investor Age Group Distribution and SIP Amount by Age Group

4.3 Fund & Market Findings

- All 5 selected bluechip funds showed consistent NAV growth from 2022-2026 with corrections during 2022 bear market
- Banking and IT sectors dominate equity fund portfolios — funds carry concentration risk in these two sectors

- Correlation matrix shows moderate-to-high correlation among equity funds — true diversification requires different asset classes
- Lumpsum transactions account for highest total amount despite SIP having highest transaction count

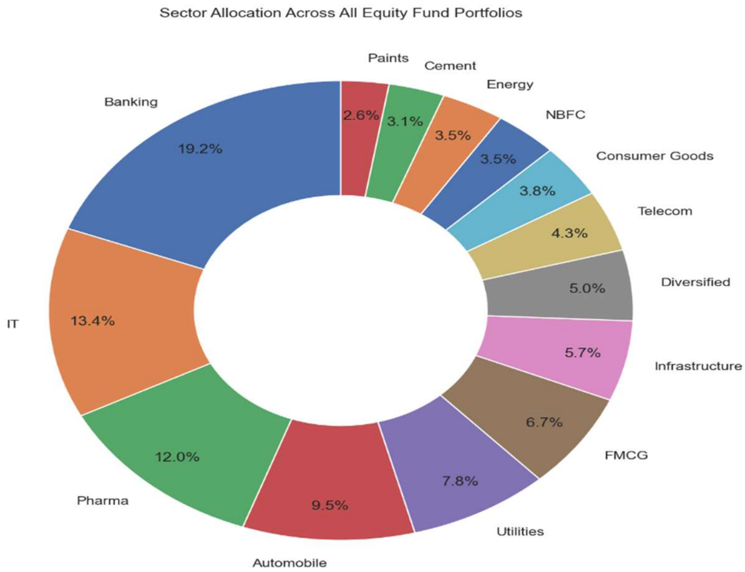

Fig 5 : Sector Allocation Across Equity Fund Portfolios

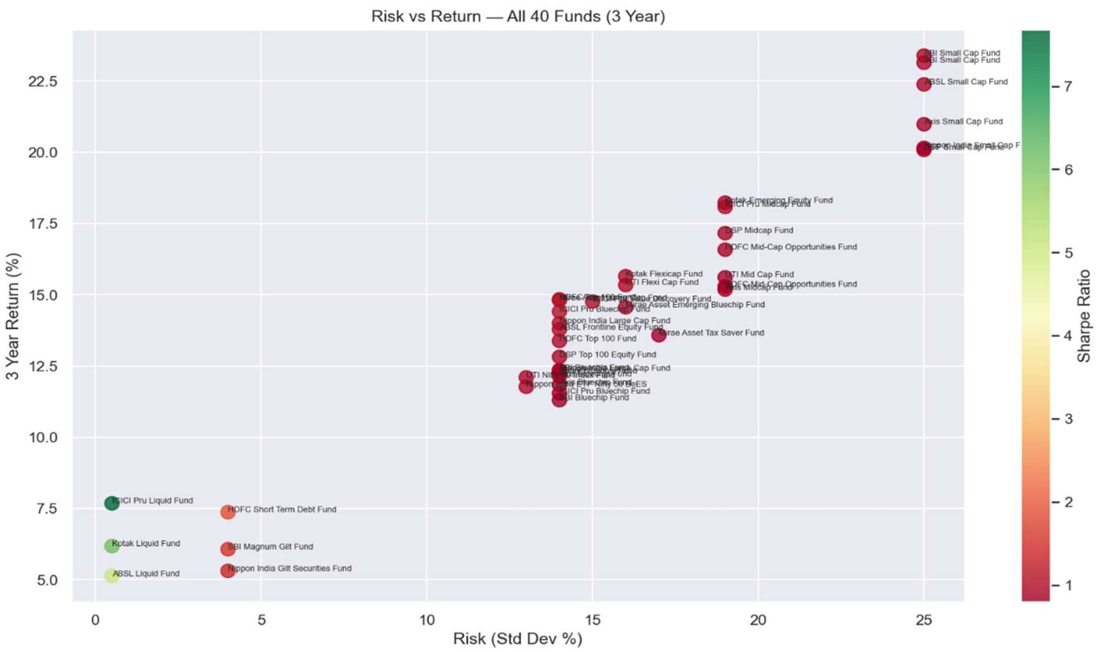

Fig 6 : Risk vs Return Scatter — All 40 Funds

4.4 Charts Created

Table 3 : Charts Created in EDA Analysis

Chart	Type	Key Insight
NAV Trend - 5 Bluechip Funds	Line chart	Consistent growth with 2022 correction
AUM by Fund House	Bar chart	SBI dominates with Rs. 12.5L Cr
Monthly SIP Inflows	Line chart	All-time high Rs. 31,002 Cr (Dec 2025)
Category Inflows Heatmap	Heatmap	Equity funds receive most consistent inflows
Investor Age Distribution	Pie + Boxplot	26-35 age group dominates
Transaction Amount by State	Bar chart	Punjab and Tamil Nadu lead
Folio Count Growth	Line chart	Doubled from 13.26 to 26.12 crore
NAV Returns Correlation	Heatmap	Moderate correlation across equity funds
Sector Allocation	Donut chart	Banking and IT dominate (>40%)
Risk vs Return Scatter	Scatter	Clear risk-return tradeoff visible

5. Fund Performance Analytics

All performance metrics computed from scratch using raw NAV history (pandas, NumPy, SciPy). Risk-free rate of 6.5% per annum used as proxy for RBI repo rate.

5.1 Metrics Computed

Table 4: Performance Metrics Computed

Metric	Formula	Purpose
CAGR	(End NAV / Start NAV)^(1/n) - 1	Annualized return over period
Sharpe Ratio	(Rp - Rf) / Std(Rp) x sqrt(252)	Risk-adjusted return (higher = better)
Sortino Ratio	(Rp - Rf) / Downside Std x sqrt(252)	Downside risk-adjusted return
Alpha	OLS intercept x 252 vs Nifty 50	Excess return over benchmark
Beta	OLS slope vs Nifty 50	Market sensitivity (1.0 = market)
Max Drawdown	min(NAV / rolling_max - 1)	Worst peak-to-trough decline
VaR (95%)	5th percentile of daily returns	Worst expected daily loss
CVaR (95%)	Mean of returns below VaR	Average loss on worst days

5.2 Top Performers by Metric

Table 5: Top Performers by Metric

Metric	Top Fund	Value
3yr CAGR	Axis Midcap Fund - Regular	36.07%
Sharpe Ratio	Mirae Asset Large Cap Fund	1.45
Sortino Ratio	Mirae Asset Large Cap Fund	Best in category
Alpha vs Nifty 50	SBI Small Cap Fund	Highest alpha
Worst Max Drawdown	SBI Small Cap Fund	-52.57%
Fund Scorecard (Composite)	ICICI Pru Midcap Fund	Score: 100/100
Worst VaR	SBI Small Cap Fund	-2.8%/day

5.3 Fund Scorecard Methodology

A composite score (0-100) was built using weighted ranking of 5 metrics:

- 30% weight: 3-year CAGR rank
- 25% weight: Sharpe Ratio rank
- 20% weight: Alpha rank (vs Nifty 50)
- 15% weight: Expense Ratio rank (inverse — lower expense = better)
- 10% weight: Max Drawdown rank (inverse — less negative = better)

5.4 Benchmark Comparison

- Top 5 scored funds all outperform Nifty 50 over the 3-year period by June 2026
- During 2022 bear market, some funds briefly underperformed Nifty 50
- By 2024-2026 all top-scored funds consistently beat the benchmark — confirming alpha generation
- This validates the fund scorecard — composite ranking successfully identifies genuinely outperforming funds

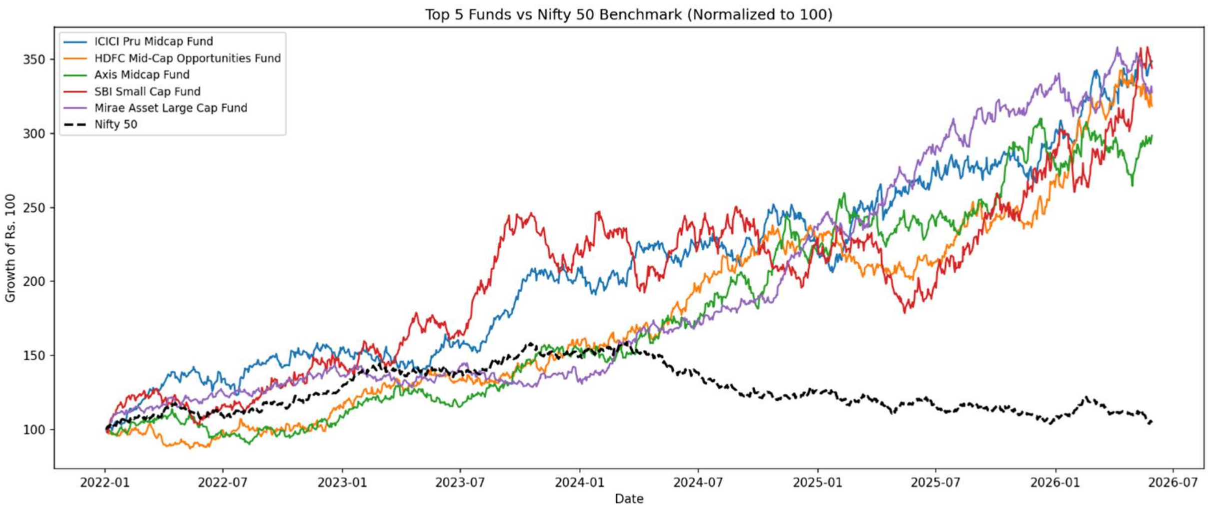

Fig 7 : Top 5 Funds vs Nifty 50 Benchmark (Normalized to 100)

6. Advanced Analytics

6.1 Value at Risk (VaR) Analysis

- Computed Historical VaR (95%) and Conditional VaR for all 40 funds from daily return distributions
- Worst VaR: SBI Small Cap Fund — on the worst 5% of days, losses exceed 2.8% daily
- Best VaR: Liquid funds — near-zero daily loss risk as expected for low-risk instruments
- CVaR confirms small cap funds carry significantly higher tail risk than large cap funds

6.2 Rolling 90-Day Sharpe Ratio

- Computed rolling Sharpe for 5 bluechip funds to assess consistency of risk-adjusted performance
- No fund consistently maintains Sharpe > 1 throughout the entire 4-year period
- All funds dip below 1 during 2022 bear market and 2024 market corrections
- Key insight: Static annual Sharpe is misleading — rolling Sharpe reveals true performance consistency

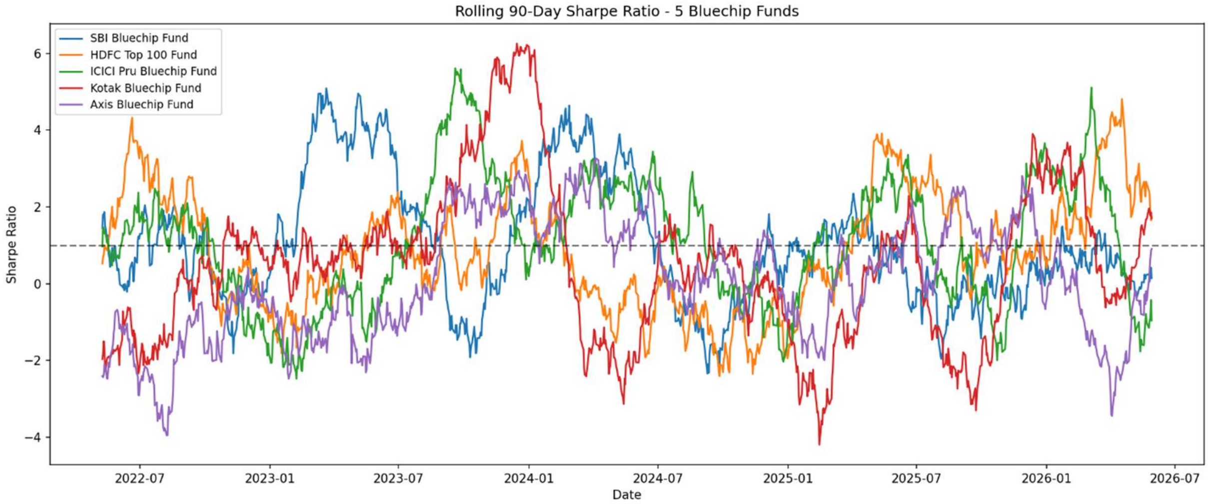

Fig 8 : Rolling 90-Day Sharpe Ratio — 5 Bluechip Funds

6.3 Investor Cohort Analysis

Table 6 : Investor Cohort Analysis

Cohort Year	Total Investors	Avg SIP Amount (Rs.)	Total Invested
2024	4,803	1,07,422	Rs. 349 crore
2025	197	1,09,158	Rs. 3 crore

- 2024 cohort dominates with 4,803 investors — platform onboarded most investors in 2024
- Average SIP amounts are remarkably similar across cohorts — consistent investor behavior
- 2025 cohort is newer hence fewer total transactions

6.4 SIP Continuity Analysis

- Analyzed all investors with 6+ SIP transactions to assess payment regularity
- 1,332 investors flagged as at-risk — average gap between SIP transactions exceeds 35 days
- These investors are at high risk of SIP discontinuation
- Recommendation: Automated reminder campaigns for investors showing gap > 30 days

6.5 Fund Recommender

Built a rule-based fund recommender that filters funds by risk appetite and ranks by Sharpe ratio:

Table 7 : Fund Recommender by Risk Appetite

Risk Appetite	Top 3 Recommended Funds	Rationale
Low	Mirae Asset Large Cap, SBI Bluechip, Nippon Large Cap	Highest Sharpe in Low/Moderate risk category
Moderate	Mirae Asset Large Cap, Mirae Asset Tax Saver, SBI Bluechip	Balance of return and risk
High	Mirae Asset Tax Saver, ICICI Pru Midcap, DSP Midcap	Best Sharpe in High/Very High risk category

6.6 Sector Concentration (HHI)

- Computed Herfindahl-Hirschman Index (HHI = sum of squared sector weights) for all equity funds
- Most concentrated fund: Axis Bluechip — heavily weighted toward Banking and IT sectors
- High HHI funds carry hidden sector concentration risk despite being marketed as diversified
- Investors in high-HHI funds have more exposure to sector-specific downturns than they may realize

7. Power BI Dashboard

A 4-page interactive Power BI dashboard was built connecting to all processed datasets. The dashboard enables self-service analytics with dynamic filtering.

Table 8 : Power BI Dashboard Pages

Page	Title	Key Visuals	Slicers
1	Industry Overview	SIP inflow line chart, AUM by fund house bar chart, KPI cards	Month
2	Fund Performance	Fund scorecard table, Top 10 funds bar chart	Category, Fund House
3	Investor Analytics	Transaction type donut, State-wise bar chart	Age Group, City Tier
4	SIP & Market Trends	SIP inflow trend, Active accounts growth, YoY growth	Year

- All 8 tables connected with relationships on amfi_code for cross-table filtering
- Dashboard exported as PDF and saved in reports/ folder
- Power BI file (.pbix) available in dashboard/ folder

8. Conclusions & Recommendations

8.1 Key Conclusions

- India's MF industry is in a structural bull run — SIP inflows, folio count, and AUM all growing consistently. Retail investor participation is at an all-time high.
- Risk-return tradeoff is confirmed — small cap funds deliver highest returns (36% CAGR) but worst drawdown (-52.57%). No fund consistently offers high return with low risk.
- Mirae Asset Large Cap is the standout fund — best Sharpe (1.45), best Sortino, top recommendation across Low and Moderate risk categories.
- SIP discontinuation is a measurable problem — 1,332 at-risk investors identified. Early intervention can significantly improve retention and AUM growth.
- B30 cities represent an untapped opportunity — similar investment capacity to T30 but significantly lower penetration.
- Sector concentration risk is underappreciated — several 'diversified' equity funds are heavily concentrated in Banking and IT.

8.2 Recommendations

Table 9 : Recommendations by Priority

Priority	Recommendation	Expected Impact
High	Automated alerts for at-risk SIP investors (gap > 35 days)	Reduce SIP discontinuation rate
High	Target B30 city investors with financial awareness campaigns	Expand investor base significantly
Medium	Promote Mirae Asset Large Cap for conservative investors	Better fund-investor risk matching
Medium	Monitor small cap fund investors during market corrections	Proactive risk management
Low	Add ML-based collaborative filtering to fund recommender	Personalized recommendations
Low	Expand HHI analysis to alert investors of concentration risk	Investor risk transparency

8.3 Limitations

- Investor transaction data is synthetically generated (real geographic and demographic distributions used)
- NAV data anchored to real historical values but projected forward using realistic return and volatility parameters
- Fund recommender is rule-based — ML-based collaborative filtering would improve personalization accuracy
- Power BI dashboard requires manual data refresh — automated pipeline would improve real-time utility

9. Technical Stack & Project Structure

Flask API: 3 endpoints built — /api/funds, /api/top-funds, /api/performance/<amfi_code>

Category	Tool / Library	Purpose
Language	Python 3.14	All data processing, ETL, analytics
Data Processing	pandas 3.0, NumPy 2.4	DataFrames, cleaning, aggregation
Statistics	SciPy 1.17	OLS regression for Alpha/Beta computation
Visualization	Matplotlib 3.7, Seaborn 0.12	15+ static charts for EDA and reports
Database	SQLite3, SQLAlchemy 2.0	Star schema database, Python-DB interface
Dashboard	Power BI Desktop	4-page interactive dashboard
Notebooks	Jupyter Lab 4.x	EDA and analytics notebooks
Version Control	Git + GitHub	Code versioning, daily commits
API	Flask 3.x, mfapi.in v1	REST API endpoints +Live NAV data (no authentication required)
HTTP Client	requests 2.30	REST API calls from Python

GitHub Repository

All code, notebooks, SQL queries, and documentation available at:

github.com/NaishadamVarshitha/bluestock-mf-capstone